Simulating the Current CT Scan Department

Stochastic Simulation — Assignment 3

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1 Introduction

The CT scan department operates two scanners that serve three types of patients: emergency patients, inpatients, and outpatients. Management wants to have a tool to analyze how well current policy works and identify problems in it.

To get a reliable way to answer these questions a discrete event simulation was developed and this report shows how it works and what results it has. Problem description section focuses on in depth problem description and current situation.Analytical capacity check section shows whether department is stable or not now. Simulation model then explains structure of a simulation. And then results are shown.

2 Problem Description

2.1 The current state

CT department currently has two CT scanners. One is opened all the time to scan emergency patients and inpatients who arrive outside office hours, the other is opened on weekdays in office hours(8.00 to 16.00). Outpatients are scheduled at the earliest on the next day after their appointment call, with 4 slots per hour in the morning and 3 per hour in the afternoon. All outpatients waiting on Friday are scheduled next week. Emergency patients are treated with priority and inpatients and outpatients are treated in order of arrival. There is no dynamic blueprint currently. Patients are placed in a waiting room with 3 chairs before scan. PAS patients are outside the scope of this report and were not included in simulation.

2.2 Performance measures

- CT scanner utilization, during and outside office hours.
- access time for outpatients.
- waiting time for emergency patients and outpatients.
- the fraction of patients who can not find a chair in waiting room.
- the percentage of inpatient requests made during office hours that are not scanned before 16:00 the same day.

Additionally for all performance indicators 95% confidence intervals will be provided with 10% relative precision.

3 Analytical Capacity Check

3.1 Analytical model

Aggregate office hour arrival rate is

$$\lambda \; = \; \lambda_E + \lambda_I + p_S \lambda_O.$$

Scan time $S \sim \text{Uniform}(10, 19)$, $E[S] = 14.5$ min and $\mu = 1/E[S]$ and load per scanner $\rho = \lambda/(c\mu)$.
Mean queuing time is $W_q = C(c, \rho)/(c\mu(1 - \rho))$, with $C(c, \rho)$ being Erlang-C delay probability.
Since service is uniform, approximation is used for $M/G/c$:

$$c_B^2 = \text{Var}(S)/E[S]^2 = 6.75/14.5^2 = 0.032$$

$$L_q^{M/G/c} \;\approx\; (1 - c_B^2)\, L_q^{M/D/c} \;+\; c_B^2\, L_q^{M/M/c},$$

3.2 Analytical results

The analytical model is stable at $c = 1$ outside office hours with utilization $\rho = 0.34$. and is stable for most of office hours at $c = 2$. However during peak inpatient requests time it temporarily exceeds 100% utilization and is unstable. But this instability is absorbed later when inpatient arrivals slow down. Overall current situation with 2 scanners is stable.

c	ρ_{avg}	$P(\text{wait})$	$E[W_q]$ M/M/c	$E[W_q]$ M/G/c
2	0.73	0.62	16.5 min	8.5 min
3	0.49	0.22	2.1 min	1.1 min

Table 1: Analytical results per scanner

4 Simulation Model

4.1 Model state

Model state is in Department object. It contains scanners, queues for each patient type, the list of pending inpatient requests, and the statistics. State changes only through events. Arrivals and completions call the start scan method that starts a scan on every non busy open scanner that can finish before its closing time.

4.2 Events

The Department model has nine event types used for a simulation. StartScan method is used throughout model events to find waiting patient and if idle scanner exists, start a scan on that patient(emergency first, then in order of arrival). And CallInpatient method is used to find pending inpatient requests and call one inpatient if there is no inpatient in transit or waiting already.

EmergencyArrival

Poisson arrivals of emergency patients.

Algorithm 1 EmergencyArrival(t)

add emergency patient to queue.
StartScan(t) of this emergency patient.
schedule next EmergencyArrival($t + \text{Exp}(60)$)

InpatientRequest

Poisson inpatient requests, rate with peaks generated by thinning.

Algorithm 2 InpatientRequest(t)

if $U(0,1) < \lambda(t)/\lambda_{\max}$ **then**
    add request and CallInpatient(t)
end if
schedule next InpatientRequest($t + \text{Exp}(1/\lambda_{\max})$)

InpatientArrival

Inpatient reaches waiting room after request.

Algorithm 3 InpatientArrival(t, request r)

clear in transit flag
add inpatient to queue
StartScan(t)

OutpatientCall

Outpatient calls arrive during weekday office hours and then are at earliest booked for next day free slots or up to Friday slots if next day is fully booked already. Otherwise outpatient goes to waiting list.

Algorithm 4 OutpatientCall(t)

book earliest free slot between next day and Friday
    if there is no free slot, caller goes to waiting list
schedule next OutpatientCall.

OutpatientAppointment

Outpatient shows at the time of booked slot with probability $p_S = 0.84$

Algorithm 5 OutpatientAppointment(t)

if $U(0, 1) < p_S$ **then** add outpatient to queue and StartScan(t)
end if

FridayPlan

Every Friday in the end of the office hours, outpatients from waiting list are scheduled next week.

Algorithm 6 FridayPlan(t)

for patient in waiting list **do** book earliest free slot next week, if any exist
end for
schedule FridayPlan next week

OpenScanner / CloseScanner

The second scanner opens only at office hours. StartScan only starts when it can finish before closing time hence it is always idle after 16.00.

Algorithm 7 OpenScanner(t) / CloseScanner(t)

update utilization accumulators at t
open scanner / close scanner

ScanCompletion

Releases scanner and scans next patient if possible.

Algorithm 8 ScanCompletion(t, scanner s)

update utilization accumulators at t
mark *scanner* idle
StartScan(t)

4.3 Data collection

Des library statistics objects inside the Department object are used to collect data.

- **Waiting times:** two SampleStatistic objects for emergency patients and outpatients recording their waiting time calculated by scan start minus arrival time.
- **Utilization:** four statistics that accumulate busy and open scanner minutes for office and outside hours.
- **Access times:** a `SampleStatistic` accumulating appointment day minus call day, recorded at booking.
- **Patients outside waiting room:** two counters, one for total arrivals, and one for patient arrivals for whom there is no chair. The measure is then their ratio.
- **Same day inpatients:** ratio of two counters, inpatient late scans by total inpatient scans requested in office hours.

First day of each replication is removed as warm up.

4.4 Verification

Consistent capacity between simulation and analytical solution. Simulation office hours utilization(0.724) and analytical solution($\rho = 0.73$) agree with each other. The same happens for outside office hours too with simulation result of 0.341.

5 Statistical Analysis

The experiment consists of $n = 100$ independent replications with different random seeds, each running for 10 weeks. First day counts as warmup and doesn't go to statistics.

For each performance indicator sample mean over $n = 100$ replications and the 95% confidence interval are gathered.

The assignment requires 10% relative precision. With $n = 100$ the largest observed deviation was 1.9%, for the outpatient waiting time and all other measures being below 1%. Therefore precision requirement is met for all indicators and bigger experiments are not needed.

6 Results

Indicator	Mean	95% CI
Utilization, office hours	0.724	[0.721, 0.726]
Utilization, outside office hours	0.341	[0.339, 0.343]
Outpatient access time (days)	1.44	[1.43, 1.44]
Emergency waiting time (min)	3.26	[3.22, 3.30]
Outpatient waiting time (min)	3.20	[3.14, 3.26]
Fraction waiting outside room	0.0032	[0.0029, 0.0035]
Inpatients not scanned same day	0.041	[0.039, 0.043]

Table 2: Current situation, $n = 100$ replications of 10 weeks, first day excluded as warm-up.

Utilization matches analytical $\rho = 0.73$. Outpatient access time is 1.4 days which can be explained by booking being possible at earliest on the next day after the call and Friday calls scheduled on Monday. Only 0.3% of patients can not find a chair in waiting room. A noticeable fraction of inpatient requests is scanned late(4.1%) driven by inpatient peak rates instability.

7 Conclusion

Current department overall is adequate and has no big structural problems. Scanners are working 73% of the time during office hours which gives very comfortable average patient waiting time of just 3 minutes. Only 0.3% of patients can not find a chair while they wait and only 4% of office hour inpatient requests are not scanned before 16.00.